3.6 A Bound on Slopes

WORKSHEET

The question

STEP I 1988, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The formula for $f'(x)$ uses only three values of f: at -1, 0 and 1. Why should three values be enough to determine the derivative of a quadratic?

STAGE 1 The formula

Covers: the formula for $f'(x)$

Ask yourself Write $f(1)$, $f(-1)$ and $f(0)$ in terms of a, b, c, substitute, and collect powers of x.

STAGE 2 The bound

Covers: $|f'(x)| \leq 4$

Ask yourself Take absolute values in the formula and use the triangle inequality, with $|f(1)|$, $|f(-1)|$, $|f(0)| \leq 1$. What bound on $|f'(x)|$ do you get, as a function of x?

Pause and try Show that $\left|x + \frac{1}{2}\right| + \left|x - \frac{1}{2}\right| + 2|x| \leq 4$ for $|x| \leq 1$, by splitting into cases according to the signs.

STAGE 3 An extreme example

Covers: the case of equality

Pause and try For the bound to be attained at $x = 1$, every inequality must be an equality. What values must $f(1)$, $f(-1)$ and $f(0)$ take? Find the quadratic through those points.

What's really going on

Markov's inequality and Chebyshev polynomials

Andrei Markov proved in 1889 that if p is a polynomial of degree n with $|p(x)| \le 1$ on $[-1, 1]$, then

$$|p'(x)| \le n^2 \qquad \text{for } |x| \le 1.$$

For $n = 2$ this is our bound of 4. The extreme cases are the *Chebyshev polynomials* T_n, defined by

$$T_n(\cos\theta) = \cos n\theta.$$

For example $\cos 2\theta = 2\cos^2\theta - 1$ gives $T_2(x) = 2x^2 - 1$, our extreme quadratic, and $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ gives $T_3(x) = 4x^3 - 3x$. Because $T_n(x) = \cos n\theta$, these polynomials oscillate between -1 and 1 as often as a polynomial of degree n possibly can, and they are as steep as possible at the ends (Figure 3.6.1). Markov's brother Vladimir extended the result to higher derivatives in 1892.

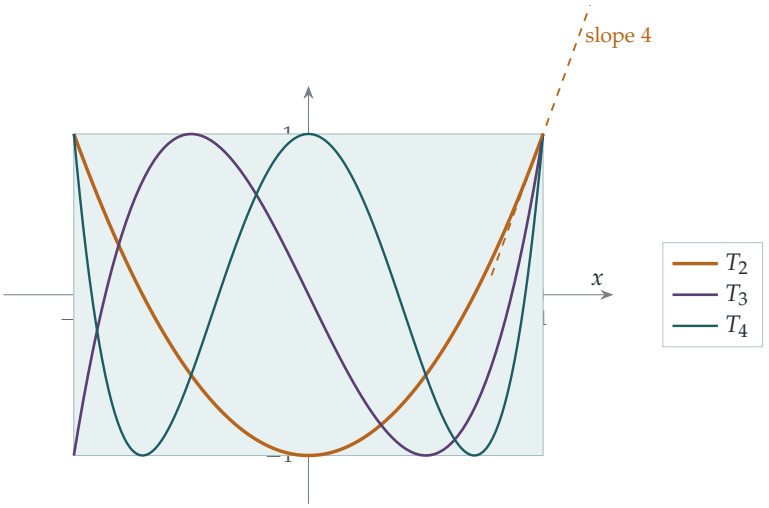


Figure 3.6.1 The Chebyshev polynomials $T_2 = 2x^2 - 1$, $T_3 = 4x^3 - 3x$ and $T_4 = 8x^4 - 8x^2 + 1$ stay inside the box $|x| \le 1$, $|y| \le 1$, touching its top and bottom as often as possible. They are the steepest polynomials of their degree at the ends: $T_n'(1) = n^2$. The dashed line is the tangent to T_2 at $x = 1$, with slope 4.

Interpolation

The formula in the question is a piece of *Lagrange interpolation*: given the values of a function at a few points, write down the unique polynomial of lowest degree through them, and use it to approximate the function or its derivative. Here the three points are $-1, 0, 1$. Formulas like this are used everywhere in numerical analysis, for example to estimate derivatives from measured data. Markov's inequality is the warning that comes with them: a small error in the values can be magnified by a factor of up to n^2 in the derivative, which is exactly the issue Mendeleev was worried about.

Chebyshev polynomials in practice

Chebyshev polynomials are the workhorses of approximation theory. Interpolating a function at the roots of T_n, rather than at equally spaced points, keeps the error under control; with equally spaced points, high-degree interpolation can go wildly wrong near the ends of the interval (Runge's phenomenon). Software that evaluates functions to full machine precision, or solves differential equations very accurately, often works with Chebyshev expansions internally.

Try this next

1 **The general idea.** Show that if f is a quadratic then $\mathrm{f}(x) = \mathrm{f}(-1)\dfrac{x(x-1)}{2} + \mathrm{f}(0)(1 - x^2) + \mathrm{f}(1)\dfrac{x(x+1)}{2}$, and differentiate to recover the formula in the question.

2 **Uniqueness.** HARDERShow that $\pm(2x^2 - 1)$ are the only quadratics with $|f(x)| \le 1$ on $[-1, 1]$ and $|f'(x)| = 4$ somewhere in $[-1, 1]$.

3 **Chebyshev polynomials.** Use $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ to check that $T_3(x) = 4x^3 - 3x$, and verify that $T_3'(1) = 9$. Show that $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$, and find T_4.

4 **Slopes of T_n.** By differentiating $T_n(\cos\theta) = \cos n\theta$ with respect to θ, show that $T_n'(\cos\theta) = \dfrac{n\sin n\theta}{\sin\theta}$, and deduce that $T_n'(1) = n^2$ by letting $\theta \to 0$.

Further reading

- **Lloyd N. Trefethen, *Approximation Theory and Approximation Practice*** (SIAM). A beautiful modern book about Chebyshev polynomials and interpolation, readable by a strong school student, with computer experiments throughout.
- **Peter Borwein & Tamás Erdélyi, *Polynomials and Polynomial Inequalities*** (Springer). For the full story of Markov's inequality and its relatives. Advanced.
- **George M. Phillips, *Interpolation and Approximation by Polynomials*** (Springer). An undergraduate text covering Lagrange interpolation, Chebyshev polynomials and Markov-type inequalities.

The full guided solution is in Section 3.6:
https://beyondstep.keithmcnulty.org/ch03/markov-bound.html